Mathematical Physics

Homework 6

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8.2.3 Show that the Chebyshev ODE, Table 7.1, may be put into self-adjoint form by multiplying by $(1-x^2)^{-1/2}$ and that this gives $w(x) = (1-x^2)^{-1/2}$ as the appropriate weighting function.

Goal: Use the weight factor $\omega(x) = (1-x^2)^{-1/2}$ to write the Chebyshev ODE in self-adjoint form.

Solution

Chebyshev ODE: $(1-x^2)y'' - xy' + n^2 y = 0$

Multiplying the ODE by $\omega(x) = (1-x^2)^{-1/2}$ yields:

$$\underbrace{\sqrt{1-x^2}}_{P_0(x)}\, y'' - \underbrace{\frac{x}{\sqrt{1-x^2}}}_{P_1(x) = P_0'(x)}\, y' + \frac{n^2}{\sqrt{1-x^2}}\, y = 0 \qquad (1)$$

Since $P_1(x) = P_0'(x)$, then ODE (1) is self-adjoint. ▨

We could have found the weight factor function by using the formula.

$$\omega(x) = P_0^{-1} \exp\left\{\int \frac{P_1(x)}{P_0(x)}\, dx\right\}$$

$$= \frac{1}{1-x^2} \exp\left\{\int \frac{-x}{1-x^2}\, dx\right\}$$

$$= \frac{1}{1-x^2} \exp\left\{\tfrac{1}{2}\ln|(1-x^2)|\right\}$$

$$= (1-x^2)^{1/2}$$ as given in the question

8.2.4 The Legendre, Chebyshev, Hermite, and Laguerre equations, given in Table 7.1, have solutions that are polynomials. Show that ranges of integration that guarantee that the Hermitian operator boundary conditions will be satisfied are

(a) Legendre $[-1, 1]$, (b) Chebyshev $[-1, 1]$,
(c) Hermite $(-\infty, \infty)$, (d) Laguerre $[0, \infty)$.

Goal: Identify the bounds of integration which satisfy the boundary conditions of self-adjoint forms of well known ODEs

Solution

For a general Hermitian operator $\mathcal{L} = \frac{d}{dx}\left[P_0 \frac{d}{dx}\right] + P_2$

$$\langle v|\mathcal{L}u\rangle = \int_a^b v^*(x)\,\mathcal{L}[u(x)]\,dx$$

$$= [P_0(v^*u' - v^{*\prime}u)]\Big|_a^b + \int_a^b \mathcal{L}[v^*(x)]\,u(x)\,dx \qquad (8.18)$$

$$= \underbrace{[P_0(v^*u' - v^{*\prime}u)]\Big|_a^b}_{\text{boundary conditions}} + \langle \mathcal{L}v|u\rangle$$

For $\mathcal{L}$ to be Hermitian, the boundary conditions must vanish at $x = a, b$.

Since solutions to the ODEs in question are polynomial, we can get the solutions to vanish at $x = a, b$ by investigating the roots of $P_0(x)$! * Arbitrary solution $u(x)$ & $v(x)$ need not necessarily vanish at boundaries.

a) Legendre:

$$(1-x^2)\frac{d^2P_n}{dx^2} - 2x\frac{dP_n}{dx} + l(l+1)P_n = 0$$

Self-adjoint form:

$$\frac{d}{dx}\left[(1-x^2)\frac{dP_n}{dx}\right] + l(l+1)P_n = 0$$

- $P_0(x) = 1 - x^2$
- Roots of $P_0(x)$ at $x = \pm 1$
- Range of Integration: $[-1, 1]$

b) Chebyshev 8

$$(1-x^2)\frac{d^2T_n}{dx^2} - x\frac{dT_n}{dx} + n^2 T_n = 0$$

Self-adjoint form:

$$\frac{d}{dx}\left[\sqrt{1-x^2}\frac{dT_n}{dx}\right] + \frac{n^2}{\sqrt{1-x^2}}T_n = 0$$

- $P_0(x) = \sqrt{1-x^2}$
- Roots of $P_0(x)$ at $x = \pm 1$
- Range of Integration: $[-1, 1]$

c) Hermite 8

$$\frac{d^2H_n}{dx^2} - 2x\frac{dH_n}{dx} + 2nH_n = 0$$

Self-adjoint form:

$$\frac{d}{dx}\left[e^{-x^2}\frac{dH_n}{dx}\right] + 2ne^{-x^2}H_n = 0$$

- $P_0(x) = e^{-x^2}$
- $P_0(x)$ decays as $x \to \pm\infty$
- Range of Integration: $(-\infty, \infty)$

b) Laguerre 8

$$xy'' + (1-x)y' + ny = 0$$

Self-adjoint form:

$$\frac{d}{dx}\left[xe^{-x}\frac{dy}{dx}\right] + ne^{-x}y = 0$$

- $P_0(x) = xe^{-x}$
- Root at $x = 0$ and decays for $x \to \infty$
- Range of Integration: $[0, \infty)$

8.3.1 Solve the Legendre equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

by direct series substitution.

(a) Verify that the indicial equation is

$$s(s-1) = 0.$$

(b) Using $s = 0$ and setting the coefficient $a_1 = 0$, obtain a series of even powers of x:

$$y_{\text{even}} = a_0\left[1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)n(n+1)(n+3)}{4!}x^4 + \cdots\right],$$

where

$$a_{j+2} = \frac{j(j+1) - n(n+1)}{(j+1)(j+2)}a_j.$$

(c) Using $s = 1$ and noting that the coefficient a_1 must be zero, develop a series of odd powers of x:

$$y_{\text{odd}} = a_0\left[x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 + \cdots\right],$$

where

$$a_{j+2} = \frac{(j+1)(j+2) - n(n+1)}{(j+2)(j+3)}a_j.$$

(d) Show that both solutions, y_{even} and y_{odd}, diverge for $x = \pm 1$ **if the series continue to infinity**. (Compare with Exercise 1.2.5.)

(e) Finally, show that by an appropriate choice of n, one series at a time may be converted into a polynomial, thereby avoiding the divergence catastrophe. In quantum mechanics this restriction of n to integral values corresponds to **quantization of angular momentum**.

Goal : Solve the Legendre Equation by the Frobenius Method

Solution

We can write $y(x) = \sum_{j=0}^{\infty} a_j x^{j+s}$ for some arbitrary $s \in \mathbb{Z}$.

Then,

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

$$(1-x^2)\sum_{j=0}^{\infty} a_j (j+s)(j+s-1) x^{j+s-2} - 2x\sum_{j=0}^{\infty} a_j (j+s) x^{s+j-1} + n(n+1)\sum_{j=0}^{\infty} a_j x^{j+s} = 0$$

$$\sum_{j=0}^{\infty} a_j \left[n(n+1) - 2(j+s) - (j+s)(j+s-1)\right] x^{j+s} + \sum_{j=0}^{\infty} a_j (j+s)(j+s-1) x^{j+s-2} = 0$$

$$\sum_{j=0}^{\infty} a_j \left[n(n+1) - (j+s)(j+s+1)\right] x^{j+s} + \sum_{j=-2}^{\infty} a_{j+2} (j+s+2)(j+s+1) x^{j+s} = 0$$

All x^{j+s} terms must vanish. Therefore,

$$a_{j+2} = \frac{(j+s)(j+s+1) - n(n+1)}{(j+s+1)(j+s+2)} a_j \quad \text{for } j \geq 0$$

$a_0\, s(s-1) = 0$ for $j = -2$

$a_1 (s+1)(s) = 0$ $\quad j = -1$

Considering $a_0 \neq 0$ yields the indicial equation $s(s-1) = 0$.

b) The case $s = 0$ and $a_1 = 0$ gives the recursive formula:

$$a_{j+2} = \frac{j(j+1) - n(n+1)}{(j+1)(j+2)} a_j$$

* $a_1 = 0$ sets all odd terms x^j to 0 effectively separating the even & odd sol'ns

c) The case $s = 1$ and $a_1 = 0$ gives the recursive formula:

$$a_{j+2} = \frac{(j+1)(j+2) - n(n+1)}{(j+2)(j+3)} a_j$$

* $a_1 = 0$ sets all odd terms of x^{j+1} to 0 recovering the even terms in x^j

d) If $x = \pm 1$, then solutions y_{odd} & y_{even} are just the series $\sum_{j=0}^{\infty} a_j$ with the relevant parity terms & sign.

Applying the Ratio Test,

$$\lim_{j\to\infty} \left| \frac{a_{j+2}}{a_j} \right| = \lim_{j\to\infty} \left| \frac{j(j+1) - n(n+1)}{(j+1)(j+2)} \right|$$

$$= \lim_{j\to\infty} \left| \frac{j^2 + O(j)}{j^2 + O(j)} \right| = 1$$

The Ratio Test is inconclusive. However, we suspect the behavior of a_j is similar to the harmonic series.

$$R_j = \frac{a_{j+2}}{a_j} = \frac{j^2 + 3j + 2 \; -2j - 2 - n(n+1)}{j^2 + 3j + 2} = 1 - \frac{2}{j} + O\left(\tfrac{1}{j^2}\right)$$

As for the harmonic series, $R_n = \frac{a_{n+2}}{a_n} = \frac{n}{n+2} = \frac{n+2-2}{n+2} = 1 - \frac{2}{n} + O\left(\tfrac{1}{n^2}\right)$

Since both series have similar ratio behaviors, their asymptotic behaviors are similar. The Harmonic Series diverges $\Rightarrow$ solutions y_{odd}, y_{even} diverge.

c) If the series terminates, then we can generate finite polynomial solutions.

$\therefore \exists j$ such that $R_j = 0$.

This happens when $j(j+1) = n(n+1)$ i.e $j=n$ for j generating even Legendre Polynomials

$j = n+1$ for j generating odd Legendre Polynomials

* Note: j is even since $a_1 = 0$

8.3.2 Show that with the weight factor $\exp(-x^2)$ and the interval $-\infty < x < \infty$ for the scalar product, the Hermite ODE eigenvalue problem is Hermitian.

Goal: Verify $w(x) = e^{-x^2}$ for a Hermite ODE

Solution

Hermite ODE: $\frac{d^2H_n}{dx^2} - 2x\frac{dH_n}{dx} + 2nH_n = 0$

$e^{-x^2}\frac{d^2H_n}{dx^2} - \underbrace{2xe^{-x^2}}_{(e^{-x^2})'}\frac{dH_n}{dx} + 2ne^{-x^2}H_n = 0$ multiplying by e^{-x^2}

$\frac{d}{dx}\left[e^{-x^2}\frac{dH_n}{dx}\right] + 2ne^{-x^2}H_n = 0$ Product Rule

Great! We got the Hermite ODE in self-adjoint form. All we need is to make sure the solutions vanish on the interval boundaries $(-\infty, \infty)$. But, we already showed this in 8.2.2!

8.3.3 (a) Develop series solutions for Hermite's differential equation

$$y'' - 2xy' + 2\alpha y = 0.$$

ANS. $s(s-1) = 0$, indicial equation.

For $s = 0$,

$$a_{j+2} = 2a_j \frac{j-\alpha}{(j+1)(j+2)} \quad (j \text{ even}),$$

$$y_{\text{even}} = a_0\left[1 + \frac{2(-\alpha)x^2}{2!} + \frac{2^2(-\alpha)(2-\alpha)x^4}{4!} + \cdots\right].$$

For $s = 1$,

$$a_{j+2} = 2a_j \frac{j+1-\alpha}{(j+2)(j+3)} \quad (j \text{ even}),$$

$$y_{\text{odd}} = a_1\left[x + \frac{2(1-\alpha)x^3}{3!} + \frac{2^2(1-\alpha)(3-\alpha)x^5}{5!} + \cdots\right].$$

(b) Show that both series solutions are convergent for all x, the ratio of successive coefficients behaving, for a large index, like the corresponding ratio in the expansion of $\exp(x^2)$.

(c) Show that by appropriate choice of α, the series solutions may be cut off and converted to finite polynomials. (These polynomials, properly normalized, become the Hermite polynomials in Section 18.1.)

a) Goal: Solve the Hermite ODE by the Frobenius Method

Solution

We can write $y(x) = \sum_{j=0}^{\infty} a_j x^{j+s}$ for some arbitrary $s \in \mathbb{Z}$.

Then,

$$y'' - 2xy' + 2\alpha y = 0$$

$$\sum_{j=0}^{\infty} a_j (j+s)(j+s-1) x^{s+j-2} - 2x \sum_{j=0}^{\infty} a_j (j+s) x^{s+j+1} + 2\alpha \sum_{j=0}^{\infty} a_j x^{s+j} = 0$$

$$\sum_{j=-2}^{\infty} a_{j+2} (j+s+1)(j+s+2) x^{s+j} - 2 \sum_{j=0}^{\infty} a_j (j+s) x^{s+j} + 2\alpha \sum_{j=0}^{\infty} a_j x^{s+j} = 0$$

All x^{j+s} terms must vanish. Therefore,

$$a_{j+2} = \frac{2(j+s) - 2\alpha}{(j+s+1)(j+s+2)} a_j \quad \text{for } j \geq 0$$

Suppose $a_0 \neq 0$

$a_0 (s-1)s = 0$ $\quad j = -2$ $\quad$ Indicial Equation: $s(s-1) = 0$

$a_1 s(s+1) = 0$ $\quad j = -1$ $\quad a_1 = 0$ to separate y_{odd} and y_{even}

b) Goal: Investigate R_n for solutions $y(x)$ and e^{-x^2}

Solution

The ratio R_j for our solutions $y(x)$ is:

$$R_j = \frac{a_{j+2}\, x^{j+2}}{a_j\, x^j} = 2\,\frac{j-\alpha}{(j+1)(j+2)}\,x^2 \quad \text{for } s=0.$$

The Ratio Test gives:

$$\lim_{j\to\infty} |R_j| = \lim_{j\to\infty} \left| 2\,\frac{j-\alpha}{(j+1)(j+2)}\,x^2 \right|$$

$$= x^2 \lim_{j\to\infty} O(1/j) = 0$$

Ratio test value of 0 for all $x \Rightarrow$ Convergence for all x

Recall the series expansion of e^{x^2}.

$$e^{x^2} = \sum_{n}^{\infty} \frac{(x^2)^n}{n!}$$

Investigating ratios R_n:

$$R_n = \frac{a_{n+1}}{a_n} = \frac{x^2}{n+1} \sim x^2\, O(1/n)$$

which is the same as the asymptotic of our solutions $y_{even}(x)$!

The odd case is analogous.

c) If the series terminates, then we can generate finite polynomial solutions.

$\therefore \exists j$ such that $R_j = 0$.

This happens when $j-\alpha=0$ i.e $j=\alpha$ for j generating even Hermite Polynomials

$j=\alpha+1$ for j generating odd Hermite Polynomials

* Note: j is even since $a_1 = 0$

8.3.4 Laguerre's ODE is

$$xL_n''(x) + (1-x)L_n'(x) + nL_n(x) = 0.$$

Develop a series solution and select the parameter n to make your series a polynomial.

Goal: Solve the Laguerre ODE by the Frobenius Method

Solution

We can write $L_n(x) = \sum_{j=0}^{\infty} a_j x^{j+s}$ for some arbitrary $s \in \mathbb{Z}$.

Then,

$$x\sum_{j=0}^{\infty} a_j (j+s)(j+s-1) x^{j+s-2} + (1-x)\sum_{j=0}^{\infty} a_j (j+s) x^{j+s-1} + n\sum_{j=0}^{\infty} a_j x^{j+s} = 0$$

$$\sum_{j=0}^{\infty} a_j \left[(j+s)(j+s-1) + (j+s) \right] x^{j+s-1} + \sum_{j=0}^{\infty} a_j \left[n-(j+s) \right] x^{j+s} = 0$$

$$\sum_{j=-1}^{\infty} a_{j+1} (j+s+1)^2 x^{j+s} + \sum_{j=0}^{\infty} a_j \left[n-(j+s) \right] x^{j+s} = 0$$

All x^{j+s} terms must vanish. Therefore,

$$a_{j+1} = \frac{j+s-n}{(j+s+1)^2} a_j \quad \text{for } j \geq 0$$

$a_0 s^2 = 0$ $\quad j=-1$ Indicial Equation: $s=0$

For $s=0$,

$$a_{j+1} = \frac{j-n}{(j+1)^2} a_j$$

$L_n(x)$ converges since $R_j \to 0$ as $j \to \infty$ $\forall x \in [0,\infty)$

In order for the series to terminate, we consider all $n=j$.

$$a_j = -\frac{n-(j-1)}{j!} a_{j-1} = (-1)^2 \frac{[n-(j-1)][n-(j-2)]}{j^2 (j-1)^2} a_{j-2} = \dots (-1)^j \frac{j[n-(j-1)][n-(j-2)]\dots n}{j^2 (j-1)^2 \dots 1^2} a_0 = \frac{(-1)^j n!/(n-j)!}{(j!)^2} a_0$$

For arbitrary n, $L_n(x) = \sum_{j=0}^{J} \frac{(-1)^j n!}{(n-j)!\, j!\, j!} a_0 x^j$ $\quad * \sum a_j x^j$